

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Business Administration (MBA)

Semester – II University Examination May 2010

MB750: Operations Management (OM)

Date: 21 May 2010, Friday

Time: 12.30 PM to 03.30 PM

Maximum Marks: 70

Instructions:

1. This paper contains six questions, of which the last one is a case analysis.
2. All questions are compulsory.
3. The first five questions carry 10 marks each and the case analysis carries 20 marks.

- Q1. (a) What is a 'competitive priority'? List the competitive priorities that form the basis of operations design. 03
- (b) Explain the difference between 'order qualifiers' and 'order winners' with a suitable illustration. 02
- (c) The activity list for a project is presented in the following table. 05

Activity	Immediate Predecessor (IP)	Estimated Duration (Weeks)
A	-	2
B	-	3
C	A	5
D	B	6
E	A	3
F	C,D	7
G	E,F	5
H	F	1

Draw a network for the project. What is the project duration? Identify the critical path and explain why it is called 'critical path'.

- Q2 (a) What are the distinguishing features of the following three manufacturing strategies? 06
- (i) Make-to-order strategy
 - (ii) Assemble-to-order strategy
 - (iii) Make-to-stock strategy

Explain each of the above with a suitable example.

- (b) Two different manufacturing processes are being considered for the production of a make-to-stock product in the consumer durables category. One process is highly labour-intensive, but has low initial capital requirement; the second process, on the other hand, is well-automated and calls for higher capital investment, with a substantially reduced variable production cost. The details of investment and variable

unit production cost for the two processes are given in the following table:

Process	Initial Investment (Rs.)	Variable Production Cost per unit (Rs.)
Process A	2,500,000	35,000
Process B	20,000,000	10,000

The demand for the product is estimated at 800 units per week.

Which of the two processes would you recommend? Why?

- Q3 (a) A new industrial hub is being created with heavy investment in an automobile manufacturing plant, supported by ancillary units for production of accessories and components. A budding entrepreneur has to choose between two sites in the vicinity of the hub for the location of a SPA, as he expects good business from the employees, their family members, and business visitors. The following table shows the different location factors, their weights, and scores (1=poor, 10=excellent) assigned by an advisory group for these two sites:

Location factor	Weight (in percent)	Score	
		Site A	Site B
Expected customers	25	6	7
Operating Costs	20	5	9
Scenic beauty and ambience	15	5	8
Adjoining activities and business centres	10	6	5
Accessibility	10	8	7
Land and construction cost	20	6	4

What are the weighted scores for the two sites? Which of these two sites will you recommend for the SPA? Why?

- (b) A chart showing a master production schedule is given below:

Quantity on hand 05 units	Lot size 50 units Lead Time 1 week								
	Week								
	1	2	3	4	5	6	7	8	
Forecast	20	10	40	10	0	0	30	20	
Customer order booked	30	20	5	8	0	2	0	0	
Projected on-hand inventory									
MPS									

Calculate the projected on-hand inventory and MPS (quantity to be produced) where required.

The following data were obtained by the quality control department of a manufacturing unit over a five-day period. All the figures apply to the product made on a single machine by a single operator. The sample size was 5. Two samples were taken per day. Comment on the process using $\bar{X}$ and R charts.

Sample Number	Observations				
	I	II	III	IV	V
1	10	12	13	8	9
2	7	10	8	11	9
3	11	12	9	12	10
4	10	9	8	13	11
5	8	11	11	7	7
6	11	8	8	11	10
7	10	12	13	13	9
8	10	12	12	10	12
9	12	13	11	12	10
10	10	13	7	9	12

The values for A_2 , D_4 and D_3 for a sample size of 5 are given below:
For $n=5$, $A_2=0.58$; $D_4=2.11$; and $D_3=0$

- (b) Describe the new trends in supply chain management as a result of the changing business environment.

Q5 (a) Explain the term 'inventory' and discuss different types of inventory used in manufacturing and service operations.

- (b) The following table shows the annual use (in millions of Rs.) of a group of similar items:

Item	Annual usage in millions of Rs. (A)	Present orders per year (N)
1	25,600	4
2	14,400	4
3	12,100	4
4	8,100	4
5	5,625	4
6	4,225	4
7	2,025	4
8	900	4
9	729	4
10	144	4

The purchase department procures each of the above ten items four times a year. Suggest a revised ordering policy, using Evert Welsch's approach, to reduce the average inventory holding of the above items. The total number of orders should not be changed.

Q.6 Read the following case and answer the questions given at the end of the case.

Adarsh's Retail

It had been a very busy week at Adarsh's retail store. A storm had blown through early in the week, and sales of goods had been brisk. This morning was relatively quiet, however, so Adarsh was using it as an opportunity to look over his shelves to get an idea of inventory levels. Some items had sold much less than he would have expected; others had sold out completely.

'I sure wish I could predict what will be sold each week', he mused. 'It seems like I always have too much of some things and not enough of others. I wonder if the POS system that our cooperative is considering would help me deal with this uncertainty.'

Adarsh's Retail is a member of a retail store cooperative, a group of more than 300 independently owned retail stores that banded together for greater buying power and better merchandise distribution. Many of the items carried by a typical retail store are similar. By buying these items as a group and storing them at a few centrally located distribution centres, individual stores can achieve economies of scale, allowing them to compete better with large nationwide chains. The cooperative is member owned. An annual membership fee and a service charge are applied to the cost of the items that a store purchases through the cooperative. Any revenues generated beyond the cooperative's operating costs are returned to members as a dividend.

Typically, a member store reviews inventory once a week and places orders that will bring stock back up to a target level. That level is the quantity of an item that, based on the time of year, the store owner wants to have on the shelf. Owners place orders by using a PC-based software program and a modem over a dial-up telephone connection to the cooperative's computer. The cooperative leases a fleet of trucks to deliver goods weekly to member stores from one of three distribution centres. Each geographic area receives shipments on a designated day. Surges in demand, if detected, can be met by mid-cycle orders shipped via UPS.

Target inventory levels are based on forecasts made from historical information kept in the store's inventory database. These forecasts are adjusted by the owner's past experience and information gleaned from trade journals and from listening to customers. Additionally, the cooperative makes aggregate sales data from member stores available, along with projected demand trends. The challenge for the store owner is to project weekly requirements accurately and to detect unusual demand for items that exceed inventory in time to avoid stockouts.

THE POS SYSTEM

The cooperative's directors have formulated a plan to obtain and install point-of-sale (POS) technology in members' stores. The motivation is to take advantage of technology that can allow the distribution center to know, in real time, what items are being sold in various stores. Armed with this information, the cooperative can improve its product forecasting, make better purchasing decisions, and reduce the chance that an item will be out of stock at distribution centers. Although the original plan was to make installation of the system

latory, the cooperative's directors decided that such a requirement could place an excessive burden on some of the smaller or less profitable stores. Consequently, installation will be optional.

The POS system comprises a scanning device attached to a cash register that operates with a microprocessor. This cash register will be networked to a PC so that an item's current price can be obtained for checkout and a perpetual inventory maintained. Each night, the distribution center will telephone the store's computer, which will answer and download the day's sales. At the end of the business day, the store owner can also review the day's sales and determine current inventory levels. The system will be designed to detect any items likely to sell out. The owner can tag any item, permitting an order to be placed that night (when the distribution center calls) for mid-cycle delivery.

The cost of the POS system will be borne by individual stores but, because of combined purchasing power, systems can be obtained for 40 percent less than list price. The cooperative's directors propose that each distribution center contract with an installer to do on-site equipment installation at individual stores. However, individual store owners would be allowed to have local technicians do the installation. The cost of system installation at the distribution center will be borne by an one-time assessment of all member stores, whether or not they install and use the system.

A two-day training session will be conducted at distribution centers whenever five or more store owners have installed the system and are ready to learn how to use it. Optionally, store owners can travel to the POS vendor's home office at their own expense for a two-day training session, which is to be offered once a month.

A vote has been scheduled prior to the cooperative's annual members' meeting. Members are asked to vote "yes" or "no" on the proposal, and a majority of those voting will determine the outcome.

As Adarsh completes his shelf scan and returns to his office, he thinks to himself: 'Since the new POS system will automatically track inventory, I wonder if I will still be able to get a gut feel of what is selling and what is not. There is nothing like examining the shelves like I just did and talking to customers to understand what I should be stocking. And, I wonder how much it will cost to run the system once it is installed.'

QUESTIONS

1. What is Adarsh's dilemma? What are the causes of his dilemma?
2. How will a POS system enhance the operations of Adarsh's Retail? How will it enhance the operations of the cooperative?
3. What criteria should be considered when assessing the benefits of the POS technology? What costs should be included?
4. If you were Adarsh, what will be your choice at the voting?